

1. Administrative & Study Identification

Full Study Title: A Study to Evaluate Mezigdomide in Combination With Carfilzomib and Dexamethasone (MeziKD) Versus Carfilzomib and Dexamethasone (Kd) in Participants With Relapsed or Refractory Multiple Myeloma (SUCCESSOR-2) **Short Title/Acronym:** SUCCESSOR-2 **Protocol Number:** CA057-008 **NCT Number:** NCT05552976 **Sponsor/Company Name:** Bristol Myers Squibb **Investigational Product:** Mezigdomide (CC-92480 / BMS-986348) **Phase of Development:** Phase 3 **Trial Status:** Recruiting **Medical Monitor:** Bristol Myers Squibb Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To compare the efficacy and safety of mezigdomide in combination with carfilzomib and dexamethasone (MeziKd) against carfilzomib and dexamethasone (Kd) in the treatment of patients with Relapsed or Refractory Multiple Myeloma (RRMM). **Secondary Objectives:** To evaluate overall survival (OS), overall response rate (ORR), duration of response (DOR), time to progression, time to next treatment, minimal residual disease (MRD) negativity, health-related quality of life, recommended mezigdomide dose, and plasma concentrations of mezigdomide in combination with carfilzomib and dexamethasone.

2.2 Background & Rationale Relapsed or refractory multiple myeloma remains a therapeutic challenge, as most patients ultimately relapse despite progress with existing therapies. Mezigdomide is a novel, oral investigational cereblon E3 ligase modulator (CELMoD) designed to enhance the degradation of transcription factors such as Ikaros and Aiolos, critical regulators of myeloma cell survival. This study aims to address a key unmet need for patients previously exposed to both anti-CD38 monoclonal antibodies and lenalidomide, evaluating whether the MeziKd triplet regimen can achieve more potent antimyeloma activity and clinically meaningful improvements over standard Kd therapy.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Inferential, seamless two-stage) **Control Type:** Active Comparator (Carfilzomib and Dexamethasone / Kd) **Blinding:** Open-label **Randomization:** Randomized (Stage 1 used a 3:3:3:2 ratio for dose-finding; Stage 2 uses a 3:2 ratio assigning MeziKd vs Kd)

3.2 Planned Enrollment & Duration Target **NS**: Approximately 525 participants (≥ 128 in Stage 1 and ≈ 397 in Stage 2) **Number of Sites**: Multicenter (International) **Estimated Study Start**: October 2022 **Estimated Primary Completion**: 18-Jul-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. 2. Documented diagnosis of multiple myeloma and measurable disease (e.g., M-protein ≥ 0.5 g/dL by serum protein electrophoresis [sPEP], or M-protein ≥ 200 mg/24-hour urine collection by uPEP, or for participants without measurable disease in sPEP or uPEP: serum free light chain levels > 100 mg/L [10 mg/dL] involved light chain and an abnormal κ/λ free light chain ratio). 3. ≥ 1 prior line of anti-MM therapy including lenalidomide and at least 2 cycles of an anti-CD38 monoclonal antibody (mAb) (participants intolerant of an anti-CD38 mAb who received < 2 cycles are still eligible). Note: One line can contain several phases. 4. Minimal response or better to ≥ 1 prior therapy and documented progressive disease (PD) during or after their last antineoplastic regimen.

4.2 Exclusion Criteria 1. Prior exposure to melphalan or carfilzomib. 2. Patient has previously received an allogeneic stem cell transplant (SCT) at any time, or received an autologous SCT within 12 weeks of initiating study treatment. 3. Other protocol-defined Inclusion/Exclusion criteria apply.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Melphalan (evaluated at 1.0, 0.6, or 0.3 mg in Stage 1 to select the optimal dose) in combination with carfilzomib and dexamethasone (Dexamethasone ATC Code: S03BA01; marketed under various trade names such as Ozurdex, Decadron, Dexium, etc.). **Route of Administration**: Oral (Melphalan, Dexamethasone) and Intravenous (Carfilzomib). **Duration of Treatment**: Treatment will continue until progressive disease (PD) or unacceptable toxicity.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for RRMM. **Prohibited**: Concurrent investigational anticancer therapies and prior melphalan or carfilzomib therapies.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to 0 Informed Consent, Medical History, Disease Assessment (sPEP/uPEP), Labs
Baseline	Day 0	Randomization, First Dose	Treatment

Cycles | Every 28 days | Safety/Efficacy Assessments, MeziKd Administration | | **End of Study** | At Progression | Final Safety Assessment, Survival and Post-Treatment Follow-up |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Progression-free survival (PFS) comparing the MeziKd arm to the Kd arm. **Measurement Method:** Independent Review Committee (IRC) and Investigator assessment based on International Myeloma Working Group (IMWG) criteria.

7.2 Secondary & Exploratory Endpoints - Overall Survival (OS) - Recommended Mezigdomide Dose - Plasma concentrations of Mezigdomide in Combination with Carfilzomib and Dexamethasone - Overall Response Rate (ORR) - Duration of Response (DOR) - Time to progression and Time to next treatment - Minimal Residual Disease (MRD) negativity

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous collection during clinic visits. Evaluated for specific toxicities inherent to the CELMoD combination regimen. **Serious Adverse Events (SAEs):** Standard reporting timeline (typically 24 hours). **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee (IDMC) utilized to review safety and interim efficacy data.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy assessments (PFS). Safety Set for all AE/SAE evaluations. **Sample Size Justification:** The study is structured with a two-stage design; Stage 1 (≥ 128 pts) ensures proper dose-finding, and Stage 2 (≈ 397 pts) provides appropriate statistical power to detect a clinically meaningful and statistically significant improvement in PFS. **Handling of Missing Data:** Standard censoring methods applied to time-to-event endpoints (e.g., PFS, OS).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine on-site monitoring and centralized data review. **Audit Trail:** Data entered via an electronic Case Report Form (eCRF) with stringent data clarification rules.

11. Study Identifiers & Governance

Primary ID: CA057-008 **Registry Number:** NCT05552976 **Posting ID:** 732743643247472503 **Sponsor/Lead Organization:** Bristol Myers Squibb **Study Title:** SUCCESSOR-2 **Official Regulatory Title:** A Phase 3, Two-stage, Randomized, Multicenter, Open-label Study Comparing Mezigdomide (CC-92480/BMS-986348), Carfilzomib, and Dexamethasone (MeziKD) Versus Carfilzomib and Dexamethasone (Kd) in Participants With Relapsed or Refractory Multiple Myeloma (RRMM): SUCCESSOR-2

12. Clinical Framework (Multiple Myeloma Specific)

Primary Condition: Relapsed or Refractory Multiple Myeloma (RRMM) / Neoplastic Process **Preferred UMLS Name:** Relapsed or Refractory Multiple Myeloma **SNOMED ID:** 763261000000108 (SNOMED Hierarchy: 703197003 | Multiple myeloma (disorder)) **Condition Definition:** Refractory Multiple Myeloma is a form of multiple myeloma, a cancer of plasma cells in the bone marrow, where the disease progresses during treatment or within a short period (typically 60 days) after the last dose of therapy. This indicates that the cancer has become resistant to the current or recent therapeutic regimens, making it challenging to achieve remission or control disease progression with standard treatments. **Diagnosis Threshold:** M-protein ≥ 0.5 g/dL by sPEP or ≥ 200 mg/24-hour by uPEP (or serum free light chain > 100 mg/L) **Medical Methodology:** Cereblon E3 ligase modulator (CELMoD) targeted degradation **Optimization Target:** Significant extension of Progression-Free Survival (PFS)

13. Study Procedures & Methodology

Management Pathway: Treat-to-progression paradigm **Education Protocol (HCP):** Site initiation visits and clinical investigator meetings **Education Protocol (Patient):** Trial onboarding, dosing schedules, and compliance education **Quality Audit Mechanism:** Centralized laboratory data review and independent response adjudication

14. Observation & Follow-Up Schedule

Total Duration: Estimated overall study completion is 25-Jul-2029 (for long-term OS tracking) **Onsite Visit Cadence:** Every treatment cycle (typically 28 days) **Remote Monitoring Frequency:** In accordance with the trial's specific clinical monitoring plan **Checklist Review Interval:** Every cycle for disease progression and safety checks

15. Sign-off & Approval

Role	Name	Signature	Date		:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					
Statistician	[Name]	_____	[DD-MMM-YYYY]					